

**Full Name:** \_\_\_\_\_  
**EEL 3135 (Spring 2026) – Lab #01**

**What to Submit on Canvas:**

- M-File: Modified `eel3135_lab01_skeleton.m` file (initial template found on Canvas)
- M-file: `myroot.m`
- PDF: A single published MATLAB PDF of your modified skeleton code

**Submission Guidelines:**

- **When plotting:**
  - Label all plots with `xlabel` and `ylabel` to describe axes and units (when appropriate)
- **When plotting time-domain signals:**
  - Choose a time axis that shows the primary characteristics of the filter or signal, unless otherwise specified
  - `stem` figures should have x-axes in samples and start at  $n = 0$
  - `plot` figures should have x-axes in actual time (usually seconds) and start at  $t = 0$

**Useful Functions to Know:**

- `plot`, `cos`, `function`, `xlabel`, `ylabel`, `exp`

**Question #1:** (*General MATLAB*) Download `eel3135_lab01_comment.m` from Canvas. Comment every line of the following MATLAB script found in the provided comment code file. Say what each line is doing in your comment. Explain each MATLAB line by using no more than one comment line, as done with the first line.

**Question #2:** (*Plotting Sinusoids*) Download `eel3135_lab01_skeleton.m` from Canvas. Following along with the skeleton code when answering this question and the following questions. Use MATLAB to plot the following sinusoids.

- (a) Use MATLAB to plot the signal

$$x[n] = 2 \cos((\pi/4)n + \pi/3)$$

Use the `stem` function. Assume the integers  $n$  are only defined between 0 and 7. The horizontal axis should be `radian` (corresponding to the values of  $(\pi/4)n$  that you sample. Label the vertical axis “Amplitude” and label the horizontal axis “Angle (Radians)”.

- (b) Use MATLAB to plot the signal

$$y[n] = \cos\left([(3\pi/100)n]^2\right)$$

Use the `plot` function. Assume the integers  $n$  are only defined between 0 and 99. The horizontal axis should be in `radians` (corresponding to the values of  $(3\pi/100)n$  that you sample. Label the vertical axis “Amplitude” and label the horizontal axis “Angle (Radians)”.

- (c) Use MATLAB to plot the signal

$$z[n] = 2 \cos((\pi/2)t) + 3 \cos((2\pi)t)$$

for values of  $t$  in the range  $-4 \leq t \leq 4$ . Choose a vector for  $t$  (i.e., sample values of  $t$ ) that creates a smooth-looking sinusoid in the plot. Use the `plot` function. Label the vertical axis “Amplitude” and label the horizontal axis “Time (Seconds)”.

**Question #3:** (*Finding Complex Roots*) This exercise shows you how to write a MATLAB function that calculates all the complex-valued solutions to the following polynomial:

$$z^n = a$$

where  $a$  is a complex number and  $n$  is a positive integer. The solutions are called the  $n$ th roots of the complex number  $a$ . For the special case of  $a = 1$ , the solutions are called the  $n$ th roots of unity (see Appendix A in the textbook). **Hint:** Let  $a = Ae^{j\phi}$  be the polar form of  $a$ . Then there are exactly  $n$  solutions/roots, for  $k = 0, 1, \dots, n-1$ , given by

$$z = A^{\frac{1}{n}} e^{j\frac{\phi+2\pi k}{n}} = \underbrace{\left(A^{\frac{1}{n}} e^{j\frac{\phi}{n}}\right)}_{r_1} \cdot \underbrace{\left(e^{j\frac{2\pi k}{n}}\right)}_{r_2},$$

- (a) Create a MATLAB function `r = myroots(n, a)` to find the  $n$ th roots of any complex number by completing the skeleton function provided in `myroots.m`. The function should take  $a$  and  $n$  above and return all of the roots `r`. Remember to comment your code.<sup>1</sup>
- (b) **Answer in your comments:** What purpose of the `help` command? **Hint:** After the generation of `myroots.m`, type in the command window, `>> help myroots`
- (c) Display (using the `disp` function) the 5th roots of  $-1$  and the 11th roots of  $2048j$ , using your function.

**Question #4:** (*AI for Signals*) One verbal description may correspond to many different signals, but an AI LLM model will generate something (maybe not what you intended). This question shows how to specify a signal with the same rigor as an equation, but using language.

- (a) Use the following prompt template to construct your prompt in UF's Navigator <https://chat.ai.it.ufl.edu/>. All blanks should be filled with a value of your choice, but **the frequencies / periods and amplitudes must be visually measurable in plots.**

You are acting as a MATLAB signal-processing assistant.

Generate a MATLAB script to produce a signal  $x(t)$  defined on  $t = \text{*** ADD HERE ***}$  to  $\text{*** ADD HERE ***}$  seconds, sampled at  $f_s = \text{*** ADD HERE ***}$  samples per second.

The signal must satisfy all of the following properties:

- Exactly  $\text{*** ADD HERE ***}$  distinct nonzero frequency components.
- The signal must be exactly periodic.
- The signal must have a mean of  $\text{*** ADD HERE ***}$ .

The output must:

1. Define the time vector  $t$
2. Define  $x(t)$  using sinusoidal components
3. Plot  $x(t)$  with proper axis labels and title

Add your prompt as a comment to your skeleton.

- (b) Paste your completed prompt into the AI system. Copy the returned MATLAB code into the skeleton. Include run the code in your submission. I should produce the requested plot.

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<sup>1</sup>It is a good practice to print out intermediate results in your function code for debugging when you are developing your function. However, please turn off printing of all intermediate results (if you know how) in the submitted version of your code to avoid generating an excessively long / confusing published report.